

Gabriele Todeschi – Résumé

Nationality: Italian
Date of birth: 9th February 1993
Email: todeschi@math.univ-lyon1.fr
Website: <https://gptod.github.io>
Github: [gptod](#)
Spoken languages: Italian, English, French



Current and previous positions

- 02/2026 -** Postdoctoral researcher at Université Claude Bernard Lyon 1, Lyon (France), within the ERC project [EYAWKAJKOS](#). Supervisor: [Filippo Santambrogio](#).
- 02/2025 -** Postdoctoral researcher at CERMICS, École Nationale des Ponts et Chaussées, Paris (France).
- 01/2026 Supervisor: [Virginie Ehrlacher](#).
- 02/2023 -** Postdoctoral researcher at LIGM, Université Gustave Eiffel, Paris (France), in the working
- 12/2024 group [New challenging Monge problems and their applications](#) of the [Labex Bézout](#). Supervisor: [François-Xavier Vialard](#)
- 02/2022 -** Postdoctoral researcher at ISTerre, Université Grenoble Alpes, Grenoble (France). Supervisors:
- 01/2023 [Jean-Marie Mirebeau](#) and [Ludovic Métivier](#)
- 07/2020 -** Visiting PhD student at the Departement of Mathematics of Università degli Studi di Padova,
- 08/2020 Padova (Italy).
- 2018-2021** PhD student in the [MOKAPLAN](#) team, Inria-Ceremade, Paris (France).
Fellow of the Marie Skłodowska-Curie Cofund MathInParis Doctoral Program of the Fondation Sciences Mathématiques de Paris (FSMP).

Education

- 2018-2021** PhD in Applied Mathematics, Université Paris-Dauphine - PSL, Paris (France).
Thesis: *Finite volume approximation of optimal transport and Wasserstein gradient flows*.
HAL: tel-03500566. Advisors: [Jean-David Benamou](#) (Inria, Paris), [Thomas Gallouët](#) (Inria, Paris), [Clément Cancès](#) (Inria, Lille).
- 2017-2018** Master M2 in Optimization - Université Paris-Saclay, Orsay (France). Average: 15.533/20.
Thesis: *A new approach to compute Wasserstein gradient flows using an H^{-1} linearization of the Wasserstein W_2 distance and its application to the numerical approximation of the solutions of a nonlinear drift diffusion equation using finite volumes*. Advisors: [Thomas Gallouët](#) (Inria, Paris), [Clément Cancès](#) (Inria, Lille).

- 2015-2018** Master's Degree in Mathematical Engineering - Università degli Studi di Padova, Padova (Italy). Curriculum: Mathematical Modelling for Engineering and Science. Score: 110/110 cum laude. Thesis: *The non-conforming Virtual Element Method (VEM) for the solution of the Navier-Stokes equations*. Advisors: [Mario Putti](#) (Università degli Studi di Padova), [Marco Manzini](#) (Los Alamos National Laboratory).
- 2012-2015** Bachelor's Degree in Energy Engineering - Università degli Studi di Padova, Padova (Italy). Curriculum: Electrical Engineering. Score: 110/110 cum laude. Thesis: *Discrete geometric formulation for bidimensional magnetostatic problems*. Advisor: [Paolo Bettini](#) (Università degli studi di Padova).
- 2007-2012** High School Diploma (scientific studies) - Liceo Scientifico "A. Rosmini", Rovereto (Italy). Score: 100/100.

Teaching

- 2025-2026** Teaching assistant, École Nationale des Ponts et Chaussées (France). Courses: *Calculus for engineers*; *Numerical statistics*.
- 2020-2021** Teaching assistant, Université Paris-Dauphine, Paris (France). Course: *Numerical methods for linear algebra and real univariate functions*.
- 2019-2020** Teaching assistant, Université Paris-Dauphine, Paris (France). Courses: *Analysis 3*; *Introduction to probability*.
- 2018-2019** Teaching assistant, Université Paris-Dauphine, Paris (France). Courses: *Analysis 3*; *Numerical methods for linear algebra and real univariate functions*; *Optimization*.
- 2016-2017** Teaching assistant, Università degli Studi di Padova, Padova (Italy). Courses: *Numerical methods for continuous systems, ordinary differential equations and partial differential equations*; *Programming (MATLAB/Fortran)*.

Students supervision

- 2021-2022** Internship supervisor, 3rd year Mathematics, ENS Paris-Saclay, Paris (France).

Other responsibilities

- 2023-2024** Organization of the seminar of the working group *New challenging Monge problems and their applications* of the Labex Bézout.
- 2023-2024** Organization of the conference [New Monge Problems and Applications](#), on computational optimal transport and its applications, held at the Université Gustave Eiffel on September 2023.

Fundings

- 2025** ANR JCJC. Project : *BARYFLOW, Barycentres de Wasserstein généralisés pour la discrétisation de flots de gradient*. Project leader: [Andrea Natale](#). Budget: 277.000 euros. Role: project member.

Journal publications

1. F. Léger, G. Todeschi, F.X. Vialard. *Nonnegative cross-curvature in infinite dimensions: synthetic definition and spaces of measures*. Geometric and Functional Analysis, 1-74, 2025.
2. T. O. Gallouët, A. Natale, G. Todeschi. *Metric extrapolation in the Wasserstein space*. Calculus of Variations and Partial Differential Equations, 64(5), 147, 2025.
3. G. Todeschi, L. Métivier, J.M. Mirebeau. *Unbalanced $L1$ optimal transport for vector valued measures and application to Full Waveform Inversion*. Journal of Computational Physics, 523:113657, 2025.
4. T. Gallouët, A. Natale, G. Todeschi. *From geodesic extrapolation to a variational BDF2 scheme for Wasserstein gradient flows*. Mathematics of Computation, 2024.
5. E. Facca, G. Todeschi, A. Natale, M. Benzi. *Efficient preconditioners for solving dynamical optimal transport via interior point methods*. SIAM Journal on Scientific Computing, 46(3), A1397-A1422, 2024.
6. A. Natale, G. Todeschi. *A mixed finite element discretization of dynamical optimal transport*. Journal of Scientific Computing, 91(2), 1-26, 2022.
7. A. Natale, G. Todeschi. *Computation of optimal transport with finite volumes*. ESAIM: Mathematical Modelling and Numerical Analysis, 55(5):1847-1871, 2021.
8. C. Cancès, T. O. Gallouët, G. Todeschi. *A variational finite volume scheme for Wasserstein gradient flows*. Numerische Mathematik, 146(3), pp. 437-480, 2020.

Conference proceedings

1. A. Natale, G. Todeschi. *TPFA finite volume approximation of Wasserstein gradient flows*. FVCA9 - International Conference on Finite Volumes for Complex Applications IX, 2020, Bergen, Norway.

Articles in preparation

1. P-C. Aubin-Frankowski, V. Ehrlacher, G. Todeschi. *Debiasing optimal transport: classical and entropic*.
2. T. O. Gallouët, A. Natale, G. Todeschi. *Extrapolating from N to 1 measures in the Wasserstein space and applications*.

Talks

1. *Metric extrapolation and signed Wasserstein barycenters*. [NEWOT](#) workshop, Laboratoire de Mathématiques d'Orsay, 2025, Orsay, France.
2. *Synthetic nonnegative cross-curvature and spaces of measures*. [PGMOdays](#) conference, EDF Lab Paris-Saclay, 2025, Palaiseau, France.
3. *Generalized Wasserstein barycenters*. [SAMM](#) seminar, Université Paris 1 Panthéon-Sorbonne, 2025, Paris, France.
4. *Metric extrapolation in the Wasserstein space*. Seminar of the [Analyse Appliquée](#) team of the Institut de Mathématiques de Marseille, 2024, Marseille, France.

5. *EVI flows via Nonnegative cross-curvature*. [Finite Volumes and Optimal Transport](#), Laboratoire de Mathématiques d'Orsay, 2024, Orsay, France.
6. *Metric extrapolation in the Wasserstein space*. Seminar of the [EDP](#) team of the Laboratoire Jean Kuntzmann, 2024, Grenoble, France.
7. *L1 optimal transport for vector valued measures and application to Full Waveform Inversion*. Seminar of the [ANEDP](#) team of the Laboratoire Paul Painlevé, 2024, Lille, France.
8. *L1 optimal transport for vector valued measures and application to Full Waveform Inversion*. [6th Dolomites Workshop on Constructive Approximation and Applications](#), 2024, Val di Fassa, Italy
9. *Extrapolation in the Wasserstein space*. [Journées SMAI-MODE](#), 2024, Lyon, France.
10. *L1 optimal transport for vector valued measures and application to Full Waveform Inversion*. Seminar GT [Transport Optimal - EDP - Machine Learning](#), 2023, Orsay, France.
11. *Optimal transport for SPD valued measures: a relaxed Beckmann's problem and application to Full Waveform Inversion*. [SEISCOPE](#) annual meeting, 2023, Grenoble, France.
12. *A relaxed Beckmann's problem for SPD valued measures and application to Full Waveform Inversion*. [Optimal Transport Theory and Applications to Physics](#), École de Physique des Houches, 2023, Les Houches, France.
13. *Application of Optimal Transport to the computation of Wasserstein gradient flows and an inverse problem in seismic imaging*. Seminar of the [EDP](#) team of the Laboratoire Jean Kuntzmann, 2023, Grenoble, France.
14. *From geodesic extrapolation to a variational BDF2 scheme for Wasserstein gradient flows*. Seminar GTCaVa ([Groupe de Travail de Calcul des Variations](#)), Université Paris-Dauphine, 2022, Paris, France.
15. *Computation of optimal transport with finite volumes*. Journées MAGA (annual meeting of the ANR project [MAGA](#)), 2022, Autrans, France.
16. *TPFA finite volume approximation of Wasserstein gradient flows*. [FVCA9 - International Conference on Finite Volumes for Complex Applications IX](#), 2020, Bergen, Norway.

Softwares

1. [W2extrapolation](#) (Fortran/Python): computation of the so-called *metric extrapolation* in the Wasserstein space between point clouds, using weak optimal transport and entropic regularization.
2. Contribution to the Fortran software [TOY2DAC](#) developed by the [SEISCOPE](#) team for seismic imaging via Full Waveform Inversion in the acoustic regime. Specifically, design of new misfit functions based on optimal transport.
3. [OT-FV](#) (Matlab): computation of dynamical quadratic optimal transport with finite volumes, owing to an interior point method for the solution of the discrete optimization problem and efficient iterative linear solvers for the saddle point problems that arise from it.
4. [dynamic-ot](#) (Python): solution of dynamical quadratic optimal transport with mixed finite element discretizations and the implementation of proximal-splitting type optimization solvers.
5. [LJKO](#), [LJKO_IPM](#) (Matlab): reliable and efficient computation of Wasserstein gradient flows via variational finite volume schemes.